Why Token Pruning Crashes Some Robot Policies but Not Others: Vision Connectors and VLA Token Compressibility

Anonymous Author(s)
Affiliation
Address
email

Abstract: Removing 5% of vision tokens collapses SmolVLA from 59.2% to 0.0% success rate, while the same reduction on $\pi_0.5$ retains 98.2% under importance-based selection but only 69.4% under random selection. Token pruning, well-established for Vision Transformers, transfers unpredictably across Vision-Language-Action (VLA) models because the vision connector governs token compressibility. We analyze this sensitivity along two axes: connector coupling (token-independent vs. token-entangling) and task complexity (single-step vs. multi-step manipulation). The coefficient of variation (CV) of post-connector token L2 norms, ranging from 0.038 to 0.314 across models, serves as a single-pass diagnostic that correlates with both pruning tolerance and method sensitivity across all three connector families. Token-entangling connectors exhibit task-amplified collapse: X-VLA loses just 1.4 percentage points on single-step Object at 90% retention but 27.8 percentage points on multi-step Goal tasks (both under random pruning), a $20\times$ task-complexity amplification. Low-CV models show $15\times$ greater sensitivity to pruning method choice than higher-CV models, and at aggressive compression random selection outperforms L2 norm ranking by 36 percentage points for low-CV architectures. For pixel-shuffle connectors where all pruning strategies yield 0.0%, Vision Token Re-encoder (VTR), a lightweight cross-attention module, recovers over 60% success rate at 75% compression through learned re-encoding. Evaluation spans six VLA architectures (450M to 7B parameters), three connector families, and four LIBERO task suites.

Keywords: vision-language-action models, token pruning, robot learning

1 Introduction

Remove 5% of vision tokens from SmolVLA and its success rate collapses from 59.2% to 0.0% across all task suites and pruning methods. Apply the same 5% reduction to $\pi_0.5$ and it retains 98.2% under L2 norm selection. Both are Vision-Language-Action (VLA) models built on a Vision Transformer (ViT) encoder feeding tokens through a connector into a language model backbone; the critical difference is not model scale (450M vs. 4B parameters) but the connector architecture. Token pruning is well-established for ViTs [1, 2, 3] and a growing body of work extends it to VLAs [4, 5, 6, 7], but each method validates on a single architecture.

These methods implicitly assume that pruning transfers across VLA designs. Yet current VLAs employ fundamentally different vision connectors: linear projections [8], MLP projectors [9], pixel-shuffle folding [10], and learned encoders with cross-token self-attention [11]. Each imposes distinct constraints on how visual information distributes across tokens. Deploying a pruning method validated on one connector type to another risks silent degradation or complete failure. No prior work evaluates pruning robustness across both connector architectures and task complexity.

Submitted to the 10th Conference on Robot Learning (CoRL 2026). Do not distribute.

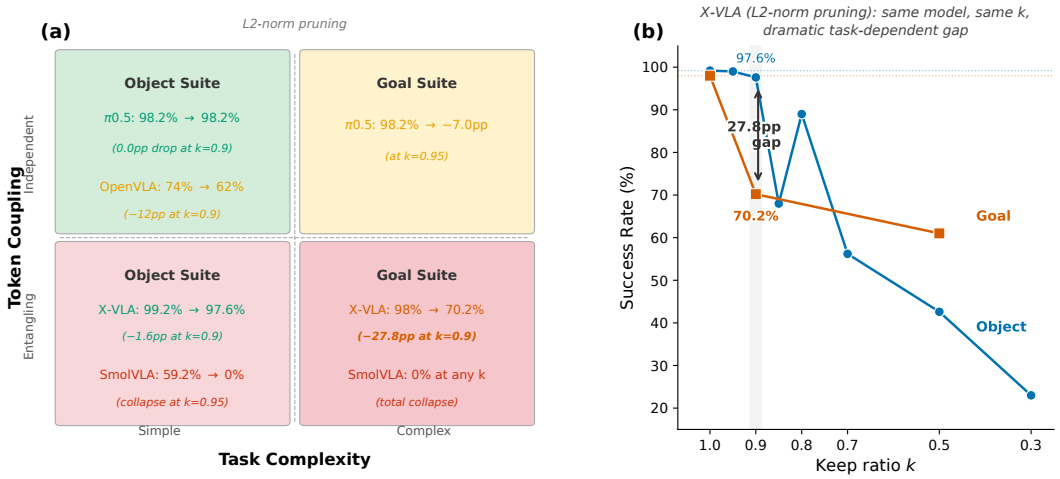


Figure 1: Two-axis framework for VLA token compressibility. Pruning robustness depends on both connector coupling (token-independent vs. token-entangling) and task complexity (single-step vs. multi-step). X-VLA loses 1.4 pp on Object at k=0.9 but 27.8 pp on Goal (random pruning), a 20× task-complexity amplification.

We propose a two-axis analysis of VLA token compressibility (Figure 1). Along the coupling axis, token-independent connectors (linear projections, MLP projectors) degrade gracefully, while token-entangling connectors produce qualitatively different failures: SmolVLA's pixel-shuffle collapses to 0.0% at every retention ratio, while X-VLA's BART encoder exhibits sharp threshold drops. Along the task complexity axis, under random pruning at k=0.9, X-VLA loses just 1.4 percentage points on single-step Object but 27.8 percentage points on multi-step Goal, a 20× amplification driven entirely by task complexity. The coefficient of variation (CV) of post-connector token L2 norms captures connector coupling on a continuous spectrum (0.038 to 0.314 across six architectures spanning all three connector families) and correlates with both pruning tolerance and method sensitivity: at k=0.95, $\pi_0.5$ (CV = 0.055) exhibits a 15× greater method gap than X-VLA (CV = 0.151).

For pixel-shuffle connectors where all pruning strategies yield 0.0%, we introduce Vision Token Re-encoder (VTR), a lightweight cross-attention module (∼22M parameters, ∼5% of SmolVLA) that recovers over 60% success rate at 75% compression through learned re-encoding.

Our contributions are:

- A two-axis compressibility analysis across six VLA architectures (450M to 7B) on four LIBERO [12] task suites, revealing connector architecture and task complexity as interacting determinants of pruning robustness.
- CV of token L2 norms as a single-pass diagnostic that correlates with pruning tolerance and method sensitivity across six architectures spanning three connector families, enabling connector-appropriate compression strategy selection without downstream evaluation.
- Vision Token Re-encoder (VTR), which recovers a pixel-shuffle VLA from 0.0% to over 60% success rate at 75% compression, where all hard pruning methods fail.

2 Related Work

ViT token pruning. DynamicViT [1], EViT [3], A-ViT [13], and Token Merging [2] remove 50–66% of tokens with under 0.5 pp ImageNet accuracy loss. Their success rests on classification's insensitivity to spatial layout; robotic manipulation inverts this assumption, as grasping depends on spatial arrangement. None has been evaluated in closed-loop VLA control.

Token efficiency in VLMs and VLAs. FastV [14] and PruMerge [15] reduce vision tokens in VLMs using language-backbone attention signals. VLA-specific methods have emerged in parallel: VLA-IAP [4] aligns pruning with action-prediction importance, VLA-Cache [5] compresses at the KV-cache level, EfficientVLA [6] combines pruning with merging, and VLA-Pruner [7] learns a pruning policy jointly with the action head. Each validates on a single architecture; no study compares pruning robustness across connector designs.

Vision connectors. The connector between ViT and language backbone governs how spatial information distributes across tokens. LLaVA [16] projects patches independently through an MLP; BLIP-2 [17] uses cross-attention queries; pixel shuffle [18] folds spatial neighborhoods into channels, adopted by SmolVLA [10]. These create qualitatively different token dependencies; independent connectors leave patch information separable, while entangling connectors distribute content across tokens. The pruning consequences have not been characterized.

We provide the first cross-architecture analysis of how connector coupling and task complexity jointly govern VLA token compressibility, introducing CV of token norms as a single-pass diagnostic that correlates with compressibility without closed-loop evaluation.

3 Method

3.1 Models and Connectors

We evaluate token pruning across six VLA architectures spanning three vision connector families (Table 1). These connectors differ in whether they process ViT patches independently or create cross-token dependencies, a distinction we formalize as *coupling*.

Linear projection applies a single matrix multiplication per patch. π_0 [8] projects 256 patches through a linear layer; $\pi_0.5$ [19] uses a $\times\sqrt{d}$-scaled linear projection across 768 tokens from three cameras (CV < 0.06).

MLP projector maps each patch independently through a multi-layer perceptron. OpenVLA [9] uses a three-layer MLP (256 tokens; CV $= 0.215$); OFT [20], an OpenVLA fine-tuning variant evaluated offline, shares the same connector (512 tokens; CV $= 0.219$).

Token-entangling connectors introduce cross-token dependencies. X-VLA [11] feeds linearly projected tokens through a 12-layer BART encoder [21] (50 tokens). SmolVLA [10] applies pixel-shuffle 4×4 folding, concatenating 16 contiguous patches into each output token's channel dimension (128 tokens).

We classify coupling as *independent* when each output token depends on a single input patch, and *entangling* when output tokens encode information from multiple patches. Four models ($\pi_0.5$, X-VLA, OpenVLA, SmolVLA) are evaluated in closed-loop control on LIBERO [12]; π_0 and OFT are analyzed via offline metrics only.

3.2 Pruning Protocol

We apply post-connector token pruning at retention ratio $k \in \{0.3, 0.5, 0.7, 0.85, 0.9, 0.95\}$, where k denotes the fraction of N vision tokens retained. Given N post-connector tokens, pruning produces $\lfloor k \cdot N \rfloor$ tokens that are passed to the LLM backbone.

Two strategies bracket expected performance:

L2 norm selection ranks tokens by their L2 norm and retains the top-k fraction. This importance-based strategy provides a favorable test: if a genuine importance hierarchy exists among tokens, L2 selection should outperform random baselines.

Random subsampling selects $\lfloor k \cdot N \rfloor$ tokens uniformly at random. Because it ignores token content entirely, random selection provides a conservative lower bound.

Table 1: VLA models evaluated in this study. Coupling indicates whether the connector processes patches independently or creates cross-token dependencies. N = number of vision tokens input to the LLM backbone. Models are grouped by evaluation protocol and ordered by CV within each group.

Model	Params	Connector	N	CV	Coupling	Eval
$\pi_0.5$	4B	Linear ($\times\sqrt{d}$)	768	0.055	Indep.	Closed-loop
X-VLA	880M	Linear + BART	50	0.151	Entangling	Closed-loop
OpenVLA	7B	3-layer MLP	256	0.215	Indep.	Closed-loop
SmolVLA	450M	Pixel-shuffle 4×4	128	0.314	Entangling	Closed-loop
π_0	3.3B	Linear	256	0.038	Indep.	Offline
OFT†	7B	3-layer MLP	512	0.219	Indep.	Offline

† OpenVLA fine-tuning variant, included for CV calibration (offline metrics only).

Both strategies physically remove unselected tokens and re-index position IDs to close gaps in the sequence. Pruned tokens are discarded, not replaced with zero vectors or mask tokens.

3.3 Compressibility Diagnostic: Importance CV

We quantify the heterogeneity of post-connector token representations using the coefficient of variation (CV) of their L2 norms:

$$\text{CV} = \frac{\sigma\big(\{\|t_i\|_2\}_{i=1}^{N}\big)}{\mu\big(\{\|t_i\|_2\}_{i=1}^{N}\big)} \tag{1}$$

where t_i denotes the i-th post-connector vision token, and σ, μ are the standard deviation and mean computed over all N token norms from a single observation.

CV is a single-pass diagnostic requiring one forward pass through the vision encoder and connector, with no downstream evaluation. The metric captures a continuous spectrum: π_0 (CV = 0.038) produces near-uniform norms, SmolVLA (CV = 0.314) exhibits high heterogeneity from pixel-shuffle, and the remaining models span the interval (Table 1). CV correlates with both pruning tolerance and sensitivity to method choice: low-CV models have near-uniform tokens where *which* tokens are removed matters more than *how many*; high-CV entangling connectors fail catastrophically under any removal strategy. We report success rate (SR) as the primary closed-loop metric throughout.

3.4 Vision Token Re-encoder (VTR)

Hard pruning fails for token-entangling connectors because removing any token destroys spatially distributed information that cannot be recovered from the remaining tokens. Vision Token Re-encoder (VTR) addresses this through learned re-encoding: a cross-attention module ($\sim$22M parameters, $\sim$5% of SmolVLA) uses M learned queries to attend over all N source tokens, aggregating information from the entire spatial field into M compressed representations. The compression ratio is controlled by $M \in \{16, 32, 64, 96, 128\}$; $M{=}N{=}128$ serves as a passthrough control. A parameter-matched MLP adapter (average pooling + two-layer MLP, no cross-attention) isolates whether gains stem from cross-attention or from re-encoding itself.

4 Experiments

4.1 Experimental Setup

We evaluate vision token pruning across six VLA architectures spanning three connector families on the LIBERO benchmark [12]. LIBERO provides four task suites of increasing complexity: Object (object manipulation), Spatial (spatial reasoning), Goal (goal-conditioned tasks), and Long (long-horizon multi-step tasks). Each suite contains 10 tasks evaluated over 50 episodes, yielding 500 episodes per condition.

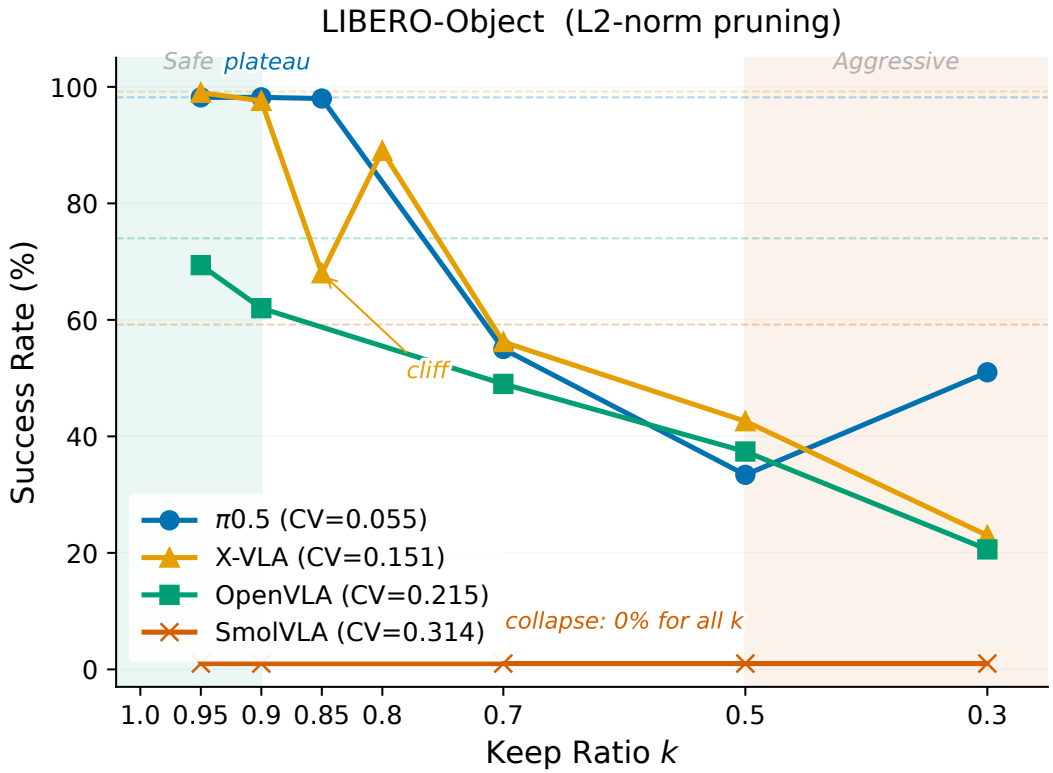


Figure 2: Pruning degradation on LIBERO-Object (L2 norm selection). Four VLAs span a complete degradation spectrum: $\pi_0.5$ maintains $\geq$98% through k=0.85, OpenVLA degrades linearly, X-VLA cliffs at k=0.7, SmolVLA collapses to 0% at k=0.95.

Four models are evaluated in closed-loop robotic control: $\pi_0.5$ [19] (4B parameters, linear projection with $\times\sqrt{d}$ scaling, 768 tokens from three cameras, CV = 0.055), X-VLA [11] (880M, linear projection followed by a 12-layer BART encoder [21], 50 tokens, CV = 0.151), OpenVLA [9] (7B, three-layer MLP projector, 256 tokens, CV = 0.215), and SmolVLA [10] (450M, pixel-shuffle 4×4 folding, 128 tokens, CV = 0.314). Two additional models are analyzed via offline metrics only: π_0 [8] (3.3B, linear projection, 256 tokens, CV = 0.038) and OFT [20] (7B, three-layer MLP, 512 tokens, CV = 0.219). All six architectures contribute to the CV diagnostic (Section 4.4). Table 1 summarizes model properties and connector details.

The pruning protocol applies post-connector token pruning at retention ratios $k \in \{0.3, 0.5, 0.7, 0.85, 0.9, 0.95\}$. Two strategies bracket the robustness range: **L2 norm selection** ranks tokens by L2 norm and retains the top-k fraction, providing an importance-based favorable test; **random subsampling** selects tokens uniformly at random, providing a conservative lower bound. Both strategies physically remove tokens and re-index position IDs. All results report mean success rate (SR) over 500 episodes with Wilson 95% confidence intervals for single-seed conditions.

4.2 Two-Axis Framework Overview

The pruning response of a VLA model depends on two axes: the degree of token independence imposed by the vision connector, and the complexity of the downstream manipulation task. Models with token-independent connectors (MLP projectors, linear projections without deep post-processing) degrade gracefully under pruning, while models with token-entangling connectors exhibit sharp, task-dependent collapse. Figure 1 illustrates this two-axis structure.

Table 2 presents closed-loop success rates across four models, four LIBERO suites, two pruning methods, and four retention ratios. The data reveal four qualitatively distinct response patterns spanning the connector spectrum (Figure 2).

$\pi_0.5$ **(linear, CV = 0.055).** On Object, L2 norm selection yields 98.2% at k=0.95, 98.2% at k=0.9, and 98.0% at k=0.85 (a plateau within 0.2 pp of the 98.2% baseline), then drops sharply to 55.0% at k=0.7. Random selection follows a different trajectory: 69.4% at k=0.95, 12.0% at k=0.9, 77.2% at k=0.7, and 69.4% at k=0.5. The low CV implies near-uniform token norms, and the method choice produces gaps as large as 86.2 pp (at k=0.9). On Spatial, $\pi_0.5$ maintains 97.8% at k=0.95 (L2) and degrades to 55.6% at k=0.5 (L2) versus 71.6% (random). The pattern is consistent: low-CV models are not pruning-immune but uniformly critical, where the choice of which tokens to remove matters more than the number removed.

X-VLA (linear + BART encoder, CV = 0.151). X-VLA exhibits threshold-graded behavior with a pronounced interaction between retention ratio and task complexity. On Object, L2 norm selection preserves 99.0% at k=0.95 and 99.2% at k=0.9 (matching the baseline), then falls sharply to 68.0% at k=0.85 with non-monotonic recovery to 89.0% at k=0.8 before continued decline to 56.2% at k=0.7. The same model at the same k=0.9 achieves only 70.2% on Goal, a 29.0 pp cross-task SR gap (99.2% − 70.2%) driven entirely by task complexity. On Long, removing just 3 of 50 tokens (k=0.95) collapses SR to 47.0% (L2) and 48.2% (random); the 1.2 pp method gap amid 47.0 pp degradation confirms the BART encoder distributes information uniformly across tokens even as it amplifies compositional task demands.

OpenVLA (MLP, CV = 0.215). OpenVLA degrades smoothly across all retention ratios on Object: 70.6% (L2) and 69.2% (random) at k=0.95, 62.0% and 63.4% at k=0.9, 49.0% and 48.5% at k=0.7, 37.4% and 35.7% at k=0.5. The near-zero method gap (within 2 pp at all retention ratios) reflects the MLP projector's per-patch independence: with a genuine importance gradient (CV = 0.215), L2 norm ranking and random selection yield comparable spatial coverage. Degradation is proportional to the fraction removed, with no threshold effects.

SmolVLA (pixel-shuffle, CV = 0.314). SmolVLA collapses to 0.0% SR at every retention ratio across all four suites and both pruning methods, including k=0.95 where only 6 of 128 tokens are removed. This catastrophic cliff is detailed in Section 4.3.

Task complexity amplification. Multi-step tasks amplify degradation in a connector-dependent manner. Under random pruning at k=0.9, X-VLA loses just 1.4 pp on Object but 27.8 pp on Goal, a 20× amplification driven entirely by task complexity. $\pi_0.5$ under L2 at k=0.95 retains 98.2% on Object but drops to 89.4% on Goal (−7.0 pp). Token-entangling connectors show sharper amplification because compositional tasks require integrating information across the cross-token dependencies that pruning disrupts. Extended cross-suite analysis is in Appendix B.

4.3 Pixel-Shuffle Entanglement

SmolVLA's pixel-shuffle connector applies 4×4 spatial folding, concatenating 16 contiguous patches into the channel dimension of a single token. This operation creates irreversible cross-token spatial dependencies: each token encodes a 4×4 receptive field, and removing any token eliminates information from all 16 constituent patches with no redundancy in the remaining tokens.

The consequence is catastrophic. SmolVLA collapses to 0.0% SR at k=0.95 (removing 6 of 128 tokens, a 5% reduction) across all four LIBERO suites (Object, Spatial, Goal, Long) and both pruning methods (L2 norm and random). We verified this result on two independent checkpoints: our LoRA fine-tuned reproduction (44.8% SR baseline) and the official SmolVLA checkpoint (59.2% SR baseline). Both exhibit identical collapse. The cliff is architectural, not a training artifact.

The pixel-shuffle mechanism explains why the collapse is method-independent. Because all tokens carry spatially entangled information from non-overlapping 4×4 grids, no scoring function can identify "dispensable" tokens. L2 norm ranking and random selection fail equally because the problem is not which tokens are removed but that any removal breaks the spatial completeness the

Table 2: Closed-loop success rates (%) on LIBERO under L2 norm (L2) and random (R) token pruning at four retention ratios. Best result per row in **bold**. SmolVLA collapses to 0.0% across all conditions. "–" indicates data not yet collected.

Model	Suite	Baseline	k=0.95 (L2 / R)	k=0.9 (L2 / R)	k=0.7 (L2 / R)	k=0.5 (L2 / R)
$\pi_0.5$	Object	**98.2**	98.2 / 69.4	**98.2** / 12.0	55.0 / 77.2	33.4 / 69.4
	Spatial	**97.2**	**97.8** / 76.4	95.0 / 54.6	67.0 / 74.4	55.6 / 71.6
	Goal	**96.4**	89.4 / 84.0	90.0 / –	– / 80.6	59.2 / 74.4
	Long	**91.6**	86.8 / 61.2	81.0 / 31.8	60.0 / 64.6	50.2 / 50.2
X-VLA	Object	**99.2**	99.0 / 98.8	**99.2** / 97.8	56.2 / 65.2	42.6 / 70.6
	Spatial	**93.0**	71.6 / 70.2	78.8 / 77.8	72.0 / –	58.8 / 71.4
	Goal	**98.4**	53.8 / 54.0	70.2 / 70.6	53.4 / 57.2	61.0 / –
	Long	**94.0**	47.0 / 48.2	27.5 / –	– / –	– / –
OpenVLA	Object	**74.0**	70.6 / 69.2	62.0 / 63.4	49.0 / 48.5	37.4 / 35.7
SmolVLA	Object	**59.2**	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0
	Spatial	**44.6**	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0
	Goal	**60.8**	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0
	Long	**36.6**	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0

downstream LLM requires. In contrast, MLP projectors (OpenVLA) and linear projections ($\pi_0.5$) map patches independently. Each token retains per-patch spatial locality, so partial removal does not destroy information from unrelated spatial positions.

4.4 When Does Pruning Method Matter? CV as Sensitivity Indicator

Token importance CV (coefficient of variation of L2 norms), measured across all six architectures spanning three connector families (Table 1), correlates with the retention ratio at which the choice of pruning method becomes critical. At mild pruning (k=0.95), method sensitivity is evaluated on the three closed-loop models with distinct CV values (excluding SmolVLA, whose 0.0% SR precludes method comparison).

For $\pi_0.5$ (CV = 0.055), the method gap at k=0.95 is substantial: 98.2% (L2) versus 69.4% (random) on Object, a 28.8 pp difference. On Spatial, the gap is 21.4 pp (97.8% vs. 76.4%). The near-uniform token norms cause L2 norm ranking to exploit small spatial correlations rather than genuine importance differences, while random selection preserves the uniform spatial coverage these models depend on.

For X-VLA (CV = 0.151), the method gap at k=0.95 is minimal on Object (0.2 pp; Table 2). Across all four suites, the method gap averages 0.8 pp (range 0.2–1.4 pp). The BART encoder produces tokens with a moderate importance gradient that both methods preserve at this mild pruning level.

For OpenVLA (CV = 0.215), the method gap at k=0.95 is 1.4 pp on Object (70.6% vs. 69.2%). The MLP projector creates a clear importance hierarchy that renders both selection strategies nearly equivalent.

The contrast is stark: $\pi_0.5$ exhibits a 28.8 pp method gap where X-VLA and OpenVLA show gaps below 1.5 pp. Low-CV models are an order of magnitude more sensitive to pruning method choice than higher-CV models at mild retention ratios. Figure 3 visualizes this relationship.

At aggressive compression (k=0.5), the method-independence observed in moderate-CV models breaks down. On Object, the gap Δ(R−L2) scales monotonically with CV: $\pi_0.5$ shows +36.0 pp (random: 69.4%, L2: 33.4%), X-VLA shows +28.0 pp (random: 70.6%, L2: 42.6%), and OpenVLA shows −1.7 pp (random: 35.7%, L2: 37.4%). At 50% retention, random selection outperforms L2 norm selection for both low-CV and moderate-CV models because random sampling preserves spatial diversity, while L2 norm ranking selects correlated high-norm tokens that cluster in similar image regions. Only OpenVLA, with CV above 0.2, produces a genuine importance gradient that L2 can exploit without spatial clustering artifacts.

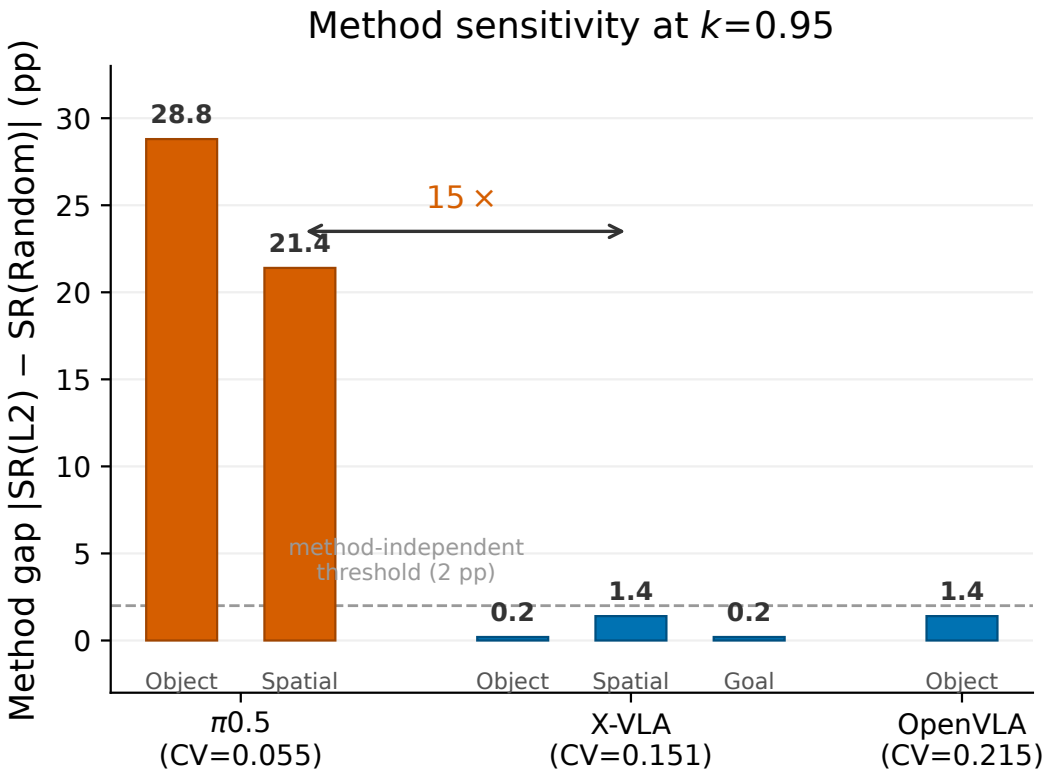


Figure 3: Method sensitivity at k=0.95. $\pi_0.5$ (CV = 0.055) exhibits method gaps of 21–29 pp, while X-VLA (CV = 0.151) and OpenVLA (CV = 0.215) remain below 1.5 pp, a 15× difference. Dashed line marks the 2 pp method-independence threshold.

4.5 VTR Recovery

Vision Token Re-encoder (VTR) addresses the catastrophic cliff through learned re-encoding rather than hard deletion. A lightweight cross-attention module (∼22M parameters, ∼5% of SmolVLA) uses M learned queries to attend over all 128 source tokens. VTR recovers SmolVLA from 0.0% to over 60% SR at 75% compression (M=32: 60.1 ± 1.3%, 3 seeds), where all hard pruning methods fail. The M-curve is flat: M=16 (87.5% compression) matches M=128 (passthrough) within 1.1 pp, confirming that learned re-encoding, not the compression ratio, is the operative factor. A parameter-matched MLP adapter achieves comparable performance (63.4 ± 2.6%, $p = 0.15$), indicating the gain stems from re-encoding itself. Cross-suite evaluation confirms generalization: Spatial 53.2% and Goal 56.4%, both recovered from 0.0%. Full results in Table 15 (Appendix F).

Limitations. All closed-loop evaluations use the LIBERO simulation benchmark; real-robot validation is absent. The models differ in backbone capacity, vision encoder, and fusion strategy beyond the connector, so the connector effect cannot be causally isolated. While CV is measured across all six architectures, the correlation between CV and closed-loop pruning tolerance relies on four models evaluated in closed-loop control. Extended discussion is in Appendix A.

5 Conclusion

Vision token pruning in VLAs is not a uniform operation: its outcome depends on the interaction between connector architecture and task complexity. Across six VLA architectures (450M–7B) and four LIBERO suites, connector coupling is the primary determinant of pruning robustness. Token-independent connectors degrade proportionally to the fraction removed; token-entangling connec-

tors exhibit task-amplified collapse. SmolVLA's pixel-shuffle produces 0.0% SR when just 5% of tokens are removed; X-VLA loses 1.4 pp on Object at k=0.9 but 27.8 pp on Goal under random pruning, a $20\times$ amplification. The CV of token L2 norms serves as a single-pass diagnostic that correlates with both pruning tolerance and method sensitivity across all three connector families. For architectures where all pruning strategies fail, VTR recovers a pixel-shuffle VLA from 0.0% to over 60% SR at 75% compression.

Two directions remain open: real-robot validation to confirm the connector-complexity interaction transfers beyond simulation, and adaptive per-task pruning schedules that adjust retention ratio to estimated task complexity.

References

[1] Y. Rao, W. Zhao, B. Liu, J. Lu, J. Zhou, and C.-J. Hsieh. DynamicViT: Efficient vision transformers with dynamic token sparsification. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[2] D. Bolya, C.-Y. Fu, X. Dai, P. Zhang, C. Feichtenhofer, and J. Hoffman. Token merging: Your ViT but faster. In *International Conference on Learning Representations (ICLR)*, 2023.

[3] Y. Liang, C. Ge, Z. Tong, Y. Song, J. Wang, and P. Xie. Not all patches are what you need: Expediting vision transformers via token reorganizations. In *International Conference on Learning Representations (ICLR)*, 2022.

[4] J. Cheng, H. Wang, W. Li, G. Wang, Y. Zhang, X. Tang, J. Wu, X. Chen, Y. Liu, and W. Zhang. VLA-IAP: Training-free visual token pruning via interaction alignment for vision-language-action models. *arXiv preprint arXiv:2603.22991*, 2026.

[5] S. Xu, Y. Wang, C. Xia, D. Zhu, T. Huang, and C. Xu. VLA-Cache: Efficient vision-language-action manipulation via adaptive token caching. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[6] Y. Yang, Y. Wang, Z. Wen, et al. EfficientVLA: Training-free acceleration and compression for vision-language-action models. *arXiv preprint arXiv:2506.10100*, 2025.

[7] Z. Liu, Y. Chen, H. Cai, et al. VLA-Pruner: Temporal-aware dual-level visual token pruning for efficient vision-language-action inference. *arXiv preprint arXiv:2511.16449*, 2025.

[8] K. Black, N. Brown, D. Driess, et al. π_0: A vision-language-action flow model for general robot control. *arXiv preprint arXiv:2410.24164*, 2024.

[9] M. J. Kim, K. Pertsch, S. Karamcheti, et al. OpenVLA: An open-source vision-language-action model. In *Proceedings of the 8th Conference on Robot Learning (CoRL)*, 2024.

[10] M. Shukor, D. Aubakirova, A. Casas, et al. SmolVLA: A vision-language-action model for affordable and efficient robotics. *arXiv preprint arXiv:2506.01844*, 2025.

[11] Y. Zheng, Z. Li, S. Wang, J. Wang, et al. X-VLA: Soft-prompted transformer as scalable cross-embodiment vision-language-action model. In *International Conference on Learning Representations (ICLR)*, 2026.

[12] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[13] H. Yin, A. Vahdat, J. M. Alvarez, A. Mallya, J. Kautz, and P. Molchanov. A-ViT: Adaptive tokens for efficient vision transformer. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[14] L. Chen et al. An image is worth 1/2 tokens after layer 2: Plug-and-play inference acceleration for large vision-language models. In *European Conference on Computer Vision (ECCV)*, 2024. Oral presentation.

[15] Y. Shang, M. Cai, B. Xu, Y. J. Lee, and Y. Yan. LLaVA-PruMerge: Adaptive token reduction for efficient large multimodal models. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.

[16] H. Liu, C. Li, Q. Wu, and Y. J. Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[17] J. Li, D. Li, S. Savarese, and S. Hoi. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International Conference on Machine Learning (ICML)*, 2023.

[18] W. Shi, J. Caballero, F. Huszar, J. Totz, A. P. Aitken, R. Bishop, D. Rueckert, and Z. Wang. Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[19] K. Black, N. Brown, N. M. M. Shafiullah, et al. $\pi_{0.5}$: a vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025.

[20] M. J. Kim, C. Finn, and P. Liang. Fine-tuning vision-language-action models: Optimizing speed and success. *arXiv preprint arXiv:2502.19645*, 2025.

[21] M. Lewis, Y. Liu, N. Goyal, M. Ghazvininejad, A. Mohamed, O. Levy, V. Stoyanov, and L. Zettlemoyer. BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.

[22] Q. Li, Y. Liang, Z. Wang, L. Luo, X. Chen, M. Liao, F. Wei, Y. Deng, S. Xu, Y. Zhang, X. Wang, B. Liu, J. Fu, J. Bao, D. Chen, Y. Shi, J. Yang, and B. Guo. Cogact: A foundational vision-language-action model for synergizing cognition and action in robotic manipulation. *arXiv preprint arXiv:2411.19650*, 2024.

A Extended Discussion

Limitations (extended). All closed-loop evaluations use the LIBERO simulation benchmark; real-robot validation is absent. The six architectures differ in backbone capacity, vision encoder, and fusion strategy beyond the connector, so the connector effect cannot be causally isolated. π_0 and OFT are evaluated through offline metrics only, without closed-loop verification. The CV diagnostic spans six architectures across all three connector families; closed-loop validation covers four models ($\pi_0.5$, X-VLA, OpenVLA, SmolVLA). SmolVLA baselines (59.2% official, 44.8% LoRA reproduction) are substantially weaker than those of the other models; whether VTR recovery generalizes to stronger baselines is unknown.

Non-monotonicity as structural property. X-VLA on Goal exhibits non-monotonic degradation: k=0.95 drops to 53.8% (L2) and 54.0% (random), while k=0.9 recovers to 70.2% under L2 selection (Table 2). The 0.2 pp method gap at k=0.95 confirms that the anomaly is independent of the token scoring criterion. This pattern is consistent with position sensitivity in the 12-layer BART encoder [21], where specific token positions carry disproportionate structural importance regardless of content.

$\pi_0.5$ **random subsampling anomaly.** Random selection on $\pi_0.5$ Object collapses from 69.4% at k=0.95 to 12.0% at k=0.9, then recovers to 77.2% at k=0.7 (Table 2). The V-shaped recovery at more aggressive pruning remains unexplained.

Small-N interactions. X-VLA operates with only 50 vision tokens. At k=0.95, removing 3 tokens produces -44.6 pp on Goal and -47.0 pp on Long (L2 norm selection). Absolute token count interacts with CV: identical retention ratios impose qualitatively different damage when the token budget is small.

Design implications. Practitioners should compute importance CV before deploying token pruning, as low-CV and entangling-connector models require distinct compression strategies. Pixel-shuffle connectors need learned re-encoding rather than hard deletion. Task complexity must factor into pruning budget decisions: retention ratios safe for simple manipulation may fail on compositional tasks.

B Cross-Suite Task Complexity Analysis

LIBERO's four suites form a complexity gradient: Object and Spatial require single-step manipulation with explicit spatial cues, while Goal and Long demand multi-step planning, goal inference, or extended temporal coordination. This gradient interacts with pruning tolerance in a connector-dependent manner.

$\pi_0.5$ shows moderate complexity amplification under L2 norm pruning at k=0.95: Object retains 98.2% (matching the baseline), Spatial retains 97.8% (within noise of the 97.2% baseline), and Goal drops to 89.4% (-7.0 pp from 96.4%). Simple tasks absorb mild pruning with negligible loss; compositional tasks expose measurable degradation even at 5% token removal.

X-VLA exhibits dramatic complexity amplification. At k=0.9 with L2 norm selection, Object retains the full 99.2% baseline, while Goal drops to 70.2% (-28.2 pp from 98.4% baseline). Under random pruning at the same ratio, Object loses just 1.4 pp (97.8%) while Goal loses 27.8 pp (70.6%), a 20$\times$ amplification. The effect is even more pronounced at k=0.95 on Long: removing just 3 of 50 tokens collapses SR from 94.0% to 47.0% (L2) and 48.2% (random), a 47.0 pp loss with only 1.2 pp method gap, while Object at the same ratio loses only 0.2 pp.

Token-entangling connectors (X-VLA's BART encoder) amplify task complexity because compositional tasks require integrating information across spatial locations. When the BART encoder creates cross-token dependencies, removing even a few tokens disrupts the distributed representations that multi-step tasks rely on. Token-independent connectors ($\pi_0.5$'s linear projection, OpenVLA's MLP) show moderate complexity amplification because each token's contribution is self-contained: removing tokens reduces spatial coverage proportionally, and simple tasks tolerate the resulting gaps.

C Pruning Protocol Details

Both pruning methods physically remove unselected tokens from the sequence, reducing the actual input length to the LLM rather than zeroing them out; zero-masking preserves positional structure while physical removal does not. After removal, remaining tokens retain their original spatial ordering; position IDs are re-indexed contiguously.

Model-specific retention ratios reflect each architecture's token count and expected sensitivity. π_0 is evaluated offline only (k=0.1, normalized L1 action loss; offline metrics only, no closed-loop evaluation available) because its deployment environment precludes closed-loop testing. $\pi_0.5$ uses $k \in \{0.3, 0.5, 0.7\}$ under both L2 and random removal; the dual-method design enables direct comparison within a single architecture. Pruning is applied per-image: each camera view independently retains the top-k fraction. X-VLA spans $k \in \{0.1, 0.2, 0.3, 0.5, 0.7, 0.8, 0.85, 0.9, 0.95\}$ with L2 removal across three seeds, supplemented by random removal at selected k values. OpenVLA uses $k \in \{0.3, 0.5, 0.7, 0.9, 0.95\}$ with random removal as the primary protocol and L2 for direct comparison. SmolVLA tests seven configurations varying scoring (learned ACIS vs. random), retention ratio ($k \in \{0.5, 0.7, 0.9, 0.95\}$), and mechanism (zero-masking, true removal, token merging via ToMe [2]).

The π_0–OpenVLA pair provides the closest available comparison: both consume 256 vision tokens from SigLIP-based encoders, but differ in connector (linear projection vs. three-layer MLP), backbone (PaliGemma vs. Llama-2), and encoder fusion strategy (single SigLIP vs. DINOv2+SigLIP). This comparison involves differences in connector, backbone, and encoder fusion, limiting causal interpretation; the primary evidence for the connector's role comes from the monotonic CV-degradation relationship across all six architectures (Section 4.4). Retention ratio k controls the fraction retained within each model, but absolute token counts vary across architectures at matched k: k=0.7 retains 35 tokens for X-VLA, 179 for OpenVLA, and 538 for $\pi_0.5$. The CV analysis derives from within-model comparisons where each model serves as its own control, eliminating this confounder.

D SmolVLA Pruning Ablation

Table 3: SmolVLA pruning ablation on LIBERO-Object. All configurations collapse to 0% SR. Two independent evaluations of the official HF checkpoint (59.2% and 57.2% baseline, within CI) confirm the cliff-edge across scoring methods.

Scoring	k	Mechanism	SR (%)	Episodes	Wilson 95% CI
Baseline (bypass)		–	44.8	500	[40.5, 49.2]
ACIS	0.50	Zero-mask	0.0	500	[0, 0.76]
ACIS	0.70	Zero-mask	0.0	500	[0, 0.76]
ACIS	0.90	Zero-mask	0.0	500	[0, 0.76]
Random	0.95	Zero-mask	0.0	500	[0, 0.76]
Random	0.50	Zero-mask	0.0	500	[0, 0.76]
ACIS	0.50	True removal	0.0	500	[0, 0.80]
ACIS	0.50	Merge (ToMe)	0.0	500	[0, 0.76]
Official checkpoint—eval protocol A (59.2% baseline)					
Baseline (bypass)		–	59.2	500	[54.8, 63.4]
Random	0.50	True removal	0.0	500	[0, 0.76]
Official checkpoint—eval protocol B (57.2% baseline)					
Baseline (bypass)		–	57.2	500	[52.9, 61.4]
Random	0.95	True removal	0.0	500	[0, 0.76]
L2 norm	0.90	True removal	0.0	500	[0, 0.76]
L2 norm	0.50	True removal	0.0	500	[0, 0.76]

E Per-Task Breakdown Tables

E.1 SmolVLA Baseline Per-Task Results

Table 4 reports per-task success rates for the SmolVLA baseline (bypass mode, no pruning) on LIBERO-Object. All pruning conditions (Table 3) produce 0% success rate on every task; we omit per-task breakdowns for pruning conditions as they carry no additional information beyond the aggregate 0% already reported.

E.2 Official SmolVLA Checkpoint Per-Task Results

Table 5 reports per-task success rates for the official SmolVLA LIBERO checkpoint (HuggingFaceVLA/smolvla_libero). Under 50% random token removal, all tasks produce 0% SR, identical to the fine-tuned checkpoint results.

Table 4: SmolVLA baseline (bypass mode) per-task success rates on LIBERO-Object (50 episodes per task, 500 total). Overall SR: 44.8%.

Task	SR (%)
alphabet_soup	22
bbq_sauce	60
butter	42
chocolate_pudding	38
cream_cheese	58
ketchup	26
milk	62
orange_juice	38
salad_dressing	50
tomato_sauce	52
Overall	**44.8**

Table 5: Official SmolVLA checkpoint per-task success rates on LIBERO-Object (eval protocol A; 50 episodes per task, 500 total). Overall SR: 59.2%. Under 50% random pruning, all tasks collapse to 0%. An independent evaluation (protocol B) yields 57.2% baseline with identical cliff-edge (Table 3).

Task	Baseline SR (%)	k=0.5 SR (%)
alphabet_soup	64	0
bbq_sauce	38	0
butter	78	0
chocolate_pudding	90	0
cream_cheese	40	0
ketchup	60	0
milk	52	0
orange_juice	80	0
salad_dressing	44	0
tomato_sauce	46	0
Overall	**59.2**	**0.0**

E.3 OpenVLA Per-Task Results Under Pruning

E.4 OpenVLA Full Degradation Curve

E.5 $\pi_0.5$ L2 Norm Multi-Seed Consistency

E.6 $\pi_0.5$ Cross-Suite Comparison

The random cliff at $k \in [0.85, 0.925]$ is task-specific: Object collapses from 69.4% (k=0.95) to 5.0% (k=0.85), whereas Spatial retains 58.4% at k=0.85 (Δ=53 pp between suites). Spatial random also exhibits mild non-monotonicity (k=0.925: 60.4% $>$ k=0.9: 54.6%, Δ=5.8 pp), but this is an order of magnitude smaller than Object's 65 pp V-dip. The cliff pattern suggests Object manipulation concentrates critical information in specific token subsets whose removal is catastrophic at narrow retention bands; Spatial tasks distribute critical information more uniformly, tolerating moderate random removal without catastrophic failure.

L2 norm selection is consistent across suites at high retention ($k \geq 0.85$): Object and Spatial differ by $\leq$ 2.4 pp (Table 9). At $k \leq 0.7$, both suites exhibit the L2 V-shape (Object valley: 33.4% at k=0.5; Spatial valley: 55.6% at k=0.5), with Spatial maintaining higher absolute SR throughout. The method preference (random $>$ L2 at k=0.5; L2 $>$ random at k=0.3) is consistent across the two suites tested, suggesting the CV-strategy interaction may extend beyond a single task distribution.

Table 6: OpenVLA per-task success rates (%) on LIBERO-Object under random token removal across five retention ratios (50 episodes per task). Baseline overall SR: 74.0%. Tasks requiring fine-grained discrimination (butter, cream_cheese) degrade earliest; visually distinctive targets (orange_juice, salad_dressing) are most resilient.

Task	k=0.95	k=0.90	k=0.70	k=0.50	k=0.30
alphabet_soup	76	70	74	60	14
bbq_sauce	38	28	46	44	30
butter	84	76	10	2	0
chocolate_pudding	54	56	32	40	10
cream_cheese	84	78	18	8	0
ketchup	82	70	68	50	30
milk	46	42	40	24	8
orange_juice	74	60	76	72	42
salad_dressing	64	54	64	36	64
tomato_sauce	92	86	62	38	8
Overall	**69.4**	**62.0**	**49.0**	**37.4**	**20.6**

Table 7: OpenVLA success rates on LIBERO-Object under random token removal at six retention ratios (50 episodes per task, 500 total per ratio). Baseline SR: 74.0%. All confidence intervals are Wilson 95%.

k	Tokens Retained	SR (%)	Wilson 95% CI
1.00 (baseline)	256	74.0	[69.9, 77.7]
0.95	243	69.4	[65.2, 73.3]
0.90	230	62.0	[57.7, 66.2]
0.70	179	49.0	[44.6, 53.4]
0.50	128	37.4	[33.3, 41.7]
0.30	77	20.6	[17.2, 24.4]

E.7 Token Importance Spatial Distribution

Token importance heatmaps reveal that $\pi_0.5$'s 768 vision tokens (reshaped to a 16×16 spatial grid after the $\sqrt{d}$-scaled linear projection) exhibit a center-high, edge-low L2 norm pattern. At k=0.9 (26 tokens removed), L2 norm-based selection discards predominantly edge tokens: 50% of removed tokens fall in the outermost ring despite edge positions comprising only 23% of the grid. This spatial concentration is consistent with the center tokens encoding the robot arm and target object, while edge tokens capture less task-relevant background.

Random removal at the same ratio samples deletions uniformly across the spatial grid, creating a combinatorial probability of simultaneously removing multiple high-importance center tokens. As more tokens are removed (k=0.925 $\rightarrow$ 0.85), this probability increases sharply, consistent with the observed cliff onset. The asymmetry explains why L2 norm selection maintains a near-baseline plateau through the cliff zone: it preferentially preserves the high-importance center region that random removal disrupts by chance.

E.8 Hypergeometric Cliff Model

The random cliff on $\pi_0.5$ Object (Section B) can be modeled as a hypergeometric sampling problem. Among N=768 vision tokens, C=91 are critical. C and T are jointly fit by minimizing RMSE against four cliff-zone data points ($k \in \{0.85, 0.9, 0.925, 0.95\}$). When $\lfloor N(1-k) \rfloor$ tokens are removed uniformly at random, the number of critical tokens removed follows a hypergeometric distribution $\mathrm{Hypergeometric}(N, C, \lfloor N(1-k) \rfloor)$. We define performance collapse as removing $\geq T$=6 critical tokens simultaneously; the fit yields C=91, T=6 with RMSE=0.025. The fitted C=91 (11.8% of tokens) is broadly consistent with the top-importance center region identified in the token heatmap (Figure 5), providing post-hoc interpretability: the "critical tokens" correspond to spatially concentrated, high-norm positions encoding the robot arm and target object.

Table 8: $\pi_0.5$ L2 norm-based removal across two seeds (500 episodes each). The V-shaped curve is reproduced with near-identical valleys (σ=0.1 pp at k=0.5). V-depth: 17.6 pp (seed42), 21.8 pp (seed43).

k	Seed 42 (%)	Seed 43 (%)	Mean $\pm$ Std	Δ (pp)
0.70	55.0	57.8	56.4 ± 2.0	+2.8
0.50	33.4	33.6	33.5 ± 0.1	+0.2
0.30	51.0	55.4	53.2 ± 3.1	+4.4

Table 9: $\pi_0.5$ pruning across LIBERO-Object (baseline 98.2%) and LIBERO-Spatial (baseline 97.2%), seed 42, 500 episodes each. Object random exhibits a deep cliff at $k \in [0.85, 0.925]$ absent on Spatial; L2 norm maintains a plateau across suites at high retention (≤ 2.4 pp difference at $k \geq 0.85$).

k	LIBERO-Object (%)		LIBERO-Spatial (%)	
	L2 Norm	Random	L2 Norm	Random
1.00	98.2	98.2	97.2	97.2
0.95	98.2	69.4	97.8	76.4
0.925	99.2	27.6	97.0	60.4
0.90	98.2	12.0	95.0	54.6
0.85	98.0	5.0	90.4	58.4
0.70	55.0	77.2	67.0	74.4
0.50	33.4	69.4	55.6	71.6
0.30	51.0	36.4	57.2	49.6

The model predicts three characteristic retention ratios: k=0.963 where $P(\text{collapse})$=10% (cliff onset), k=0.937 where $P(\text{collapse})$=50% (midpoint), and k=0.902 where $P(\text{collapse})$=90% (near-complete collapse). The cliff width $\Delta k \approx 0.06$ between 10% and 90% collapse probability matches the observed transition from 69.4% SR (k=0.95) to 12.0% SR (k=0.90). The model captures the cliff onset and sharpness but does not explain the V-shape recovery at $k \leq 0.7$, where success rate rebounds to 77.2%. The recovery lies outside the model's scope (which assumes monotonic collapse probability). We hypothesize that at aggressive pruning the model transitions to a qualitatively different computation regime, potentially relying on coarser spatial features that survive uniform subsampling, but this remains untested.

The cross-suite difference emerges naturally from the critical token count. Object manipulation concentrates critical information in C=91 tokens (11.8% of the grid), creating a steep increase in $P(\text{collapse} \geq T)$ over a narrow k range. Spatial tasks distribute critical information across approximately $C \approx 5$ tokens, making the probability of simultaneously removing $\geq T$ critical tokens negligible at all tested retention ratios, consistent with Spatial's graceful degradation (58.4% at k=0.85 vs. Object's 5.0%).

E.9 OpenVLA Multi-Seed Consistency

Table 10: OpenVLA success rates (%) on LIBERO-Object under random token removal across two seeds (500 episodes each). All values are within 5 pp, confirming the graded degradation curve is reproducible. Seed 7 is reported in the main text.

k	Seed 7 (%)	Seed 43 (%)	Δ (pp)
0.90	62.0	62.2	+0.2
0.70	49.0	44.4	−4.6
0.50	37.4	35.6	−1.8

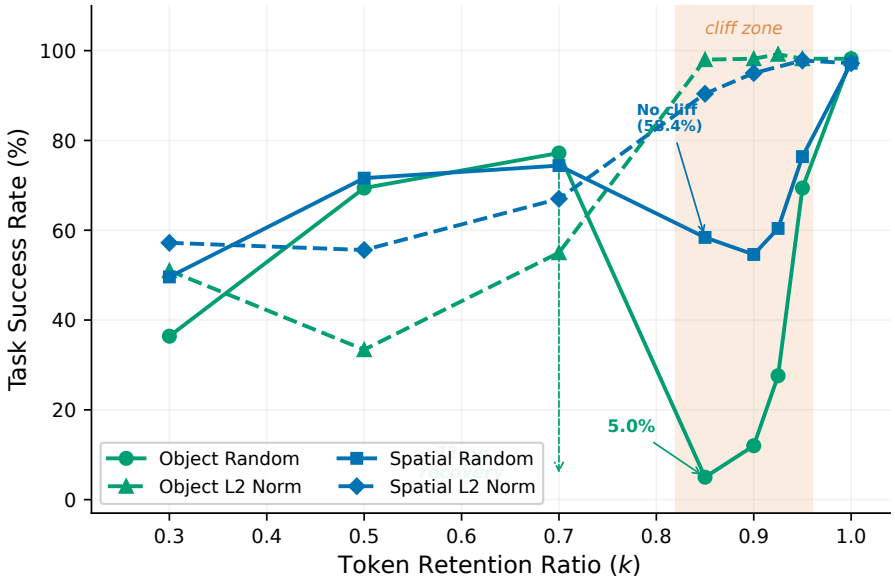


Figure 4: $\pi_0.5$ cross-suite pruning profiles. Object random (green solid) exhibits a deep cliff at $k \in [0.85, 0.925]$ with 72 pp recovery, absent on Spatial (blue solid). L2 norm (dashed) maintains a plateau on both suites through the cliff zone.

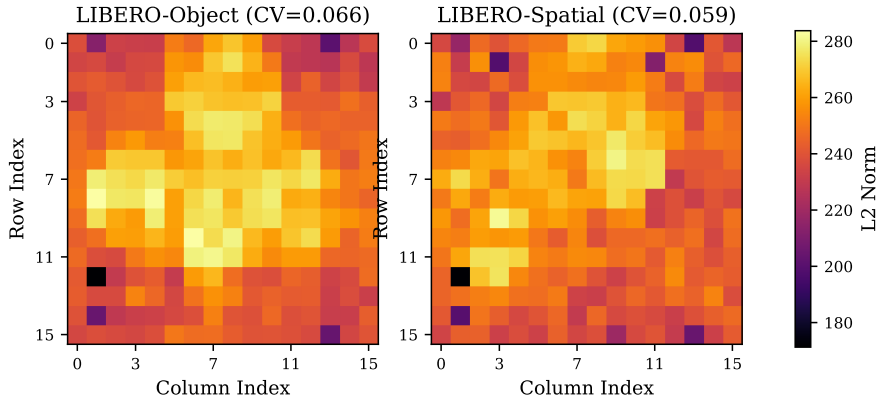


Figure 5: $\pi_0.5$ token importance spatial distribution (16×16 grid, averaged over timesteps). Left: LIBERO-Object shows a pronounced center-high, edge-low pattern (center rows 7–9: norm ∼265–267; edge rows 0, 14–15: norm ∼235–236; CV=0.066). Right: LIBERO-Spatial exhibits a flatter distribution (center-edge gap 14.67 vs. 21.87; CV=0.059). The 33% center-edge gap difference quantitatively explains Object's sharp cliff versus Spatial's graceful degradation under random removal.

E.10 L2 Norm vs. Random Scoring on OpenVLA

E.11 X-VLA Cliff Localization: Per-Task Analysis

The non-monotonic pattern is reproducible across both seeds: the same four tasks (butter, chocolate_pudding, cream_cheese, ketchup) exhibit catastrophic drops at k=0.85, while the remaining six maintain or improve performance. This task-specific sensitivity is consistent with the BART encoder creating entangled dependencies among marginal tokens: at k=0.85, two additional low-importance tokens (L2 rank 41–42) introduce interference that selectively disrupts tasks requiring fine-grained spatial discrimination. The four affected tasks all involve small, visually similar grocery items, while the robust tasks (alphabet_soup, salad_dressing) feature distinctive objects or high-contrast targets.

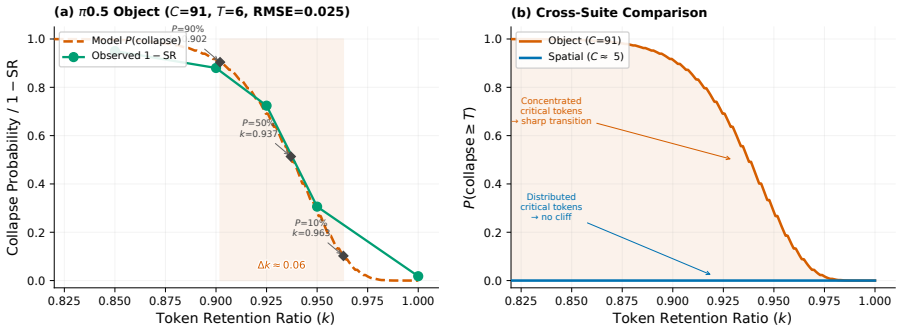


Figure 6: Hypergeometric cliff model for $\pi_0.5$ Object random pruning. (a) Model-predicted collapse probability (dashed) vs. observed success rate (solid) across the cliff zone $k \in [0.85, 0.95]$; RMSE=0.025. (b) $P(\text{collapse})$ curves for Object (C=91, concentrated) and Spatial ($C \approx 5$, distributed), illustrating why Object exhibits a sharp phase transition absent on Spatial.

Table 11: OpenVLA success rates (%) on LIBERO-Object under L2 norm-based vs. random token removal (500 episodes per configuration). Random uses seed = 7; L2 uses seed = 42. The gap widens to 4.4 pp at k=0.3 and 8.6 pp at k=0.9. Same-seed (seed42) random at k=0.3 yields 21.4% (L2 outperforms by 3.6 pp); same-seed at k=0.7 yields 44.4% (L2 outperforms by 5.4 pp). Δ is defined as L2−Random (positive indicates L2 advantage), opposite to Table 20.

k	L2 Norm SR (%)	Random SR (%)	Δ (pp)
0.95	70.6	69.4	+1.2
0.90	70.6	62.0	+8.6
0.70	49.8	49.0	+0.8
0.50	39.2	37.4	+1.8
0.30	25.0	20.6	+4.4

E.12 Smart Compress Per-Task Comparison

E.13 Smart Compress Multi-Seed Results

F VTR Full Results

G Full Smart Compress Ablation

Table 16 reports all five Smart Compress configurations. The cross-attention module improves performance regardless of compression: $M \in \{32, 64, 128\}$ all achieve ∼60% SR, exceeding the 44.8% baseline by 15–17 points ($p < 0.001$, two-proportion z-test). Only at $M = 16$ (87.5% compression) does performance decline modestly to 55.8%, still 11 points above baseline. The M=96 dip (mean 54.5% across three seeds, Table 14) may reflect an intermediate bottleneck that introduces representational redundancy without the regularization benefit of stronger compression (M=32, M=64).

Table 17 reports performance across task suites. All suites are in-distribution: the training set (HuggingFaceVLA/libero) covers Object, Spatial, Goal, and Long. On LIBERO-Spatial, Smart Compress achieves 53.2% SR (vs. 0% under hard pruning); on LIBERO-Goal, 56.4% SR (−6.6pp from the 63.0% unpruned baseline, vs. 0% under pruning). The LIBERO-Goal decline indicates that re-encoding benefits vary across task structures even within the training distribution. Per-task results are in Table 13.

Table 12: X-VLA per-task success rates (%) at k=0.80 (40 of 50 tokens) and k=0.85 (42 of 50 tokens) on LIBERO-Object. Per-task breakdown is mean of 2 seeds (50 episodes per task per seed, 100 total); overall values are confirmed across 3 seeds ($89.0 \pm 0.7\%$ at k=0.80, $67.9 \pm 0.3\%$ at k=0.85). Despite retaining *fewer* tokens, k=0.80 outperforms k=0.85 by 21.1 pp overall. Four tasks (**bold**) suffer catastrophic interference ($\Delta \leq -50$ pp) from two additional low-importance tokens (L2 rank 41–42 of 50) at k=0.85; the remaining six are unaffected ($|\Delta| \leq 10$ pp). Cross-seed variance is ≤ 4 pp per task.

Task	k=0.80 (%)	k=0.85 (%)	Δ (pp)
alphabet_soup	100.0	100.0	0
bbq_sauce	85.0	87.0	+2
butter	**57.0**	**0.0**	**−57**
chocolate_pudding	**88.0**	**35.0**	**−53**
cream_cheese	**98.0**	**45.0**	**−53**
ketchup	**79.0**	**27.0**	**−52**
milk	85.0	95.0	+10
orange_juice	96.0	98.0	+2
salad_dressing	100.0	100.0	0
tomato_sauce	98.0	91.0	−7
Overall	**89.0**	**67.9**	**−21.1**

Table 13: Smart Compress per-task success rates on LIBERO-Object for $M \in \{32, 64, 128\}$ (seed 42; 3-seed means are reported in Table 15). All three achieve similar overall SR (59.8%–61.6%), but individual task performance varies substantially across configurations, with per-task differences up to 30 percentage points.

Task	M=32 (%)	M=64 (%)	M=128 (%)
alphabet_soup	68	74	62
bbq_sauce	46	56	34
butter	82	84	82
chocolate_pudding	90	74	82
cream_cheese	64	38	58
ketchup	66	64	62
milk	76	62	56
orange_juice	28	46	58
salad_dressing	60	64	54
tomato_sauce	36	54	50
Overall	**61.6**	**61.6**	**59.8**

H Smart Compress Evaluation Across Task Suites

Table 17 reports the performance of Smart Compress (M=32) across three LIBERO suites. All suites are in-distribution: the training set (HuggingFaceVLA/libero) covers Object, Spatial, Goal, and Long (40 tasks, 1693 episodes total). On LIBERO-Goal, performance drops from 63.0% to 56.4% (−6.6pp). On LIBERO-Spatial, Smart Compress achieves 53.2% SR, where the official SmolVLA checkpoint scores 43.8% and hard pruning yields near-zero (0.6%). While 53.2% nominally exceeds the official Spatial baseline, this difference concentrates in two of ten tasks (+44pp and +48pp respectively); the remaining eight tasks show a net difference of only +0.2pp. We therefore do not treat the Spatial comparison as evidence that re-encoding improves over the uncompressed official baseline. The qualitative pattern (cliff-edge vulnerability and re-encoding recovery) extends across the three tested task structures within the training distribution. The LIBERO-Goal decline indicates that re-encoding benefits vary across task structures.

Table 14: Smart Compress success rates across three seeds on LIBERO-Object (500 episodes each). Mean ± std reported.

M	Seed 42	Seed 43	Seed 44	Mean ± Std
32	61.6	59.2	59.6	60.1 ± 1.3
64	61.6	60.0	64.4	62.0 ± 2.3
96	52.6	58.2	52.8	54.5 ± 3.2
128	59.8	54.0	60.2	58.0 ± 3.5

Table 15: VTR re-encoding on SmolVLA (LIBERO-Object). Cross-attention and MLP adapter results report mean ± std over 3 seeds (500 episodes each). The M-curve is flat: M=16 (87.5% compression) matches M=128 (passthrough) within 1.1 pp. The baseline control (100K extra training, no compressor) confirms the gain is architectural. "Compression" = fraction of tokens removed.

Configuration	M	Compression	SR (%)
SmolVLA-Base (LoRA)	128	0%	44.8
SmolVLA official	128	0%	59.2
Baseline control (+100K steps)	128	0%	45.0
Hard pruning (any k)	–	5–50%	0.0
Cross-attn VTR	16	87.5%	56.9 ± 1.3
Cross-attn VTR	32	75%	60.1 ± 1.3
Cross-attn VTR	64	50%	**62.0 ± 2.2**
Cross-attn VTR	96	25%	54.5 ± 3.2
Cross-attn VTR	128	0%	58.0 ± 3.5
MLP adapter (∼22M)	32	75%	63.4 ± 2.6

I Smart Compress Training Details

The Smart Compress module is a cross-attention pooling layer inserted between SmolVLA's pixel-shuffle vision connector and the Qwen2-0.5B language backbone. It consists of M learned query vectors that attend over the 128 vision tokens through standard multi-head cross-attention, producing M compressed tokens as input to the LLM. The compressor adds 22.2M trainable parameters (approximately 5% of SmolVLA's 451.5M total). Training is end-to-end: compressor and VLA policy parameters are jointly optimized.

The loss profile differs between configurations: $M = 32$ has higher train loss (0.11 vs. 0.045) but lower eval loss (0.052 vs. 0.062), consistent with the smaller bottleneck acting as a stronger regularizer. Despite this loss difference, both achieve identical closed-loop success rates (61.6%), reinforcing the finding that offline loss does not predict closed-loop performance (Section P).

I.1 MLP Adapter Control

To verify that re-encoding effectiveness is not specific to cross-attention, we train a parameter-matched MLP adapter as a control. The adapter is a 3-layer feed-forward network with expansion ratio 4 (hidden dimension $4 \times d_{\text{model}}$), using average pooling for token reduction from 128 to M tokens. Total parameters: 22.14M, matched to the cross-attention module's 22.2M. Training follows the identical protocol as Smart Compress: SmolVLA-Base checkpoint, 100K steps, learning rate 10^{-4}, batch size 4, same LIBERO dataset. At $M = 32$ (75% reduction), the MLP adapter achieves $63.4 \pm 2.6\%$ SR over 3 seeds, statistically indistinguishable from the cross-attention result ($60.1 \pm 1.3\%$; two-sample t-test, $p = 0.15$). This confirms that the re-encoding benefit does not depend on the specific module architecture.

Table 16: Full Smart Compress ablation on LIBERO-Object across five compression levels. M=32 and M=64 achieve similar SR (∼60%); M=96 shows a modest dip; the M=128 passthrough confirms the gain originates from re-encoding.

Method	M	Reduction	SR (%)	Episodes	Wilson 95% CI
Baseline	128	0%	44.8	500	[40.5, 49.2]
Extended training	–	–	45.0	500	[40.7, 49.3]
Hard pruning	–	50–90%	0.0	500+	[0, 0.76]
Smart Compress	128	0%	58.0	1500	±3.5
Smart Compress	96	25%	52.6	500	[48.2, 56.9]
Smart Compress	64	50%	61.6	500	[57.3, 65.8]
Smart Compress	32	75%	61.6	500	[57.3, 65.8]
Smart Compress	16	87.5%	55.8	500	[51.4, 60.1]

M=128 reports mean ± std over 3 seeds (500 episodes each). Other Smart Compress rows report single seed (seed 42) with Wilson 95% CI. See Table 14 for full multi-seed results.

Table 17: Smart Compress (M=32) evaluation across LIBERO task suites. All suites are in-distribution (training covers Object, Spatial, Goal, Long). Hard pruning (any method, k=0.50) produces 0% or near-zero on all suites.

Suite	Baseline (unpruned)	Smart Compress M=32	Hard Pruning k=0.50
Object	44.8%	61.6%	0%
Spatial	51.8%	53.2%	0.6%
Goal	63.0%	56.4%	0%

J Offline Single-Step Action Error Analysis

Table 19 reports the single-step action prediction error under pruning, computed over 200 evaluation frames from LIBERO-Object. The unpruned model (bypass mode) produces actions with an L2 distance of 0.414 from ground truth; under ACIS pruning at k=0.50, this increases to 1.391 (3.4×). The per-dimension breakdown of the pruned-vs-unpruned difference reveals that the gripper dimension dominates (L2 = 1.02), corresponding to complete gripper state inversion on many frames. Position errors (0.20–0.30 per axis) are moderate individually but compound across the 7-dimensional action space. These single-step errors are large enough to push the robot off-trajectory within a few steps, explaining why 0% of pruned episodes succeed in closed-loop evaluation despite moderate average offline metrics.

K Supplementary Figures

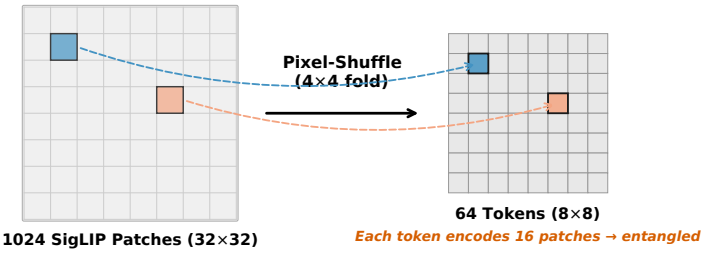


Figure 7: Pixel-shuffle folding: each 4×4 block of SigLIP patches is channel-concatenated into one token. Removing any token eliminates its receptive field (gray) with no redundancy.

Table 18: Smart Compress training hyperparameters. All configurations ($M \in \{16, 32, 64, 96, 128\}$) use identical procedures; only the number of query vectors differs.

Hyperparameter	Value
Model	
Base checkpoint	SmolVLA-Base (pretrained, before LIBERO fine-tuning)
Compressor architecture	Cross-attention, 2 layers, 12 heads, dropout 0.1
Compressor parameters	22.2M
Total trainable parameters	122.1M / 472.2M (25.85%)
Optimization	
Optimizer	AdamW (β_1=0.9, β_2=0.95, weight decay 10^{-10})
Learning rate	10^{-4}
Schedule	Linear warmup (1K steps, start factor 0.01) $\rightarrow$ cosine annealing ($\eta_{\min}$=2.5×10^{-6})
Max gradient norm	10.0
Total training steps	100,000
Data & batching	
Dataset	HuggingFaceVLA/libero (40 tasks, 1693 episodes across Object, Spatial, Goal, Long suites)
Batch size	4 (no gradient accumulation)[†]
Num workers	4
Eval interval	every 5K steps
Checkpoint interval	every 1K steps
Hardware & runtime	
Hardware	1$\times$ NVIDIA RTX 4090 D (24GB)
Precision	fp32 (no mixed precision)
Wall clock	$\sim$6h 43m per configuration
Training speed	4.1 samples/sec
Seed	42 (seeds 43, 44 for multi-seed runs)
$M = 32$ *final metrics*	
Train action loss	0.11
Eval action loss	0.052
$M = 64$ *final metrics*	
Train action loss	0.045
Eval action loss	0.062

[†]The official SmolVLA checkpoint uses batch size 64 across 8 GPUs (effective batch size 512). Our batch size 4 on a single GPU contributes to the lower baseline SR (44.8% vs. 87.3%). All Smart Compress configurations and the extended training control use this same batch size for fair comparison.

L Attention Entropy Measurement Details

We measure the attention entropy ratio as the normalized Shannon entropy of the attention weights that the action decoder assigns to vision token positions. For a model with N vision tokens, the entropy ratio is:

$$H_{\text{ratio}} = \frac{-\sum_{i=1}^{N} \alpha_i \log \alpha_i}{\log N} \tag{2}$$

where α_i is the attention weight on the i-th vision token. The ratio ranges from 0 (all attention on one token) to 1 (uniform attention).

SmolVLA. Measured on 2400 frames from LIBERO rollouts (overall entropy) and 260 frames across all 10 LIBERO-Object tasks (per-layer analysis across 16 layers). The mean per-layer entropy ratio is 0.95 (range: 0.93 to 0.98), consistent with near-uniform attention across all 128 vision tokens. The per-layer breakdown confirms this uniformity is not an averaging artifact:

- All 16 layers: 0.93–0.98 (spread 0.05)
- Minimum: layer 13 (0.93)

Table 19: Single-step action prediction error (SmolVLA, k=0.50, ACIS, zero-mask, 200 LIBERO-Object evaluation frames). Pruning increases action error by 3.4× and introduces catastrophic gripper inversions.

Metric	Unpruned	Pruned (k=0.50)
Action L2 vs. ground truth	0.414	1.391
Pruned vs. unpruned L2		1.284
Relative action error	–	111.5%
Per-dimension L2 (pruned vs. unpruned)		
Position (x, y, z)	–	0.20, 0.25, 0.30
Rotation (r_x, r_y, r_z)	–	0.01, 0.02, 0.01
Gripper	–	1.02

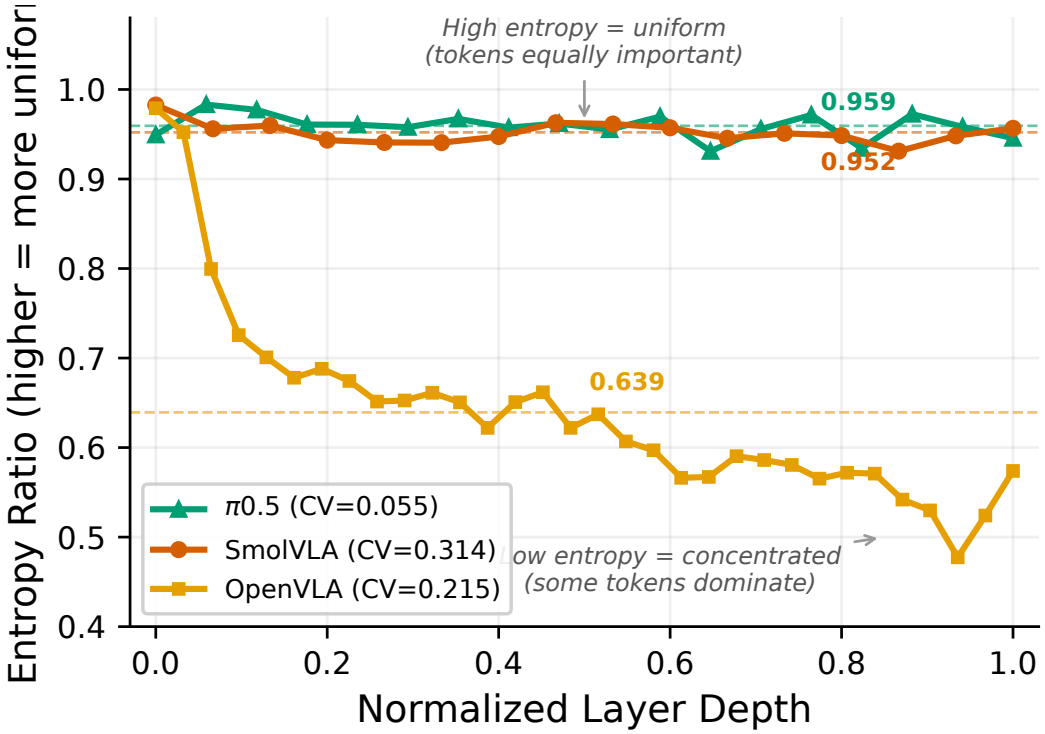


Figure 8: Per-layer attention entropy ratio for SmolVLA (16 layers), OpenVLA (32 layers), and π_0 (18 layers). SmolVLA and π_0 maintain near-uniform attention throughout, while OpenVLA progressively sharpens from 0.98 to 0.48. Despite similar entropy, SmolVLA and π_0 exhibit opposite pruning outcomes due to token importance heterogeneity (CV = 0.314 vs. 0.038). X-VLA (entropy = 0.614, CV = 0.151) occupies an intermediate position, consistent with its threshold-graded pattern (onset at k=0.85).

The flat profile across all processing depths confirms that pixel-shuffle tokens resist selective attention at every layer: because each token encodes 16 spatially contiguous patches via channel concatenation, all tokens are comparably information-dense, and the transformer has no basis for developing spatial selectivity.

π_0. Measured on 100 frames from LIBERO-Object (tasks 20–29) across all 18 layers of the Gemma-based PaliGemma backbone (512 vision tokens: 256 per image × 2 cameras). The mean suffix-to-vision entropy ratio is 0.959 (range: 0.931–0.983, std 0.013), near-uniform across all layers with no progressive sharpening:

- All 18 layers: 0.931–0.983 (spread 0.052)

- Minimum: 0.931; Maximum: 0.983

The flat entropy profile, combined with π_0's extremely low CV (0.038), is consistent with its high tolerance to pruning: tokens carry near-equal importance and receive near-equal attention. Removing any subset approximates uniform spatial downsampling, preserving a proportional share of the visual field without disrupting the attention distribution.

OpenVLA. Measured on 250 frames from real LIBERO rollouts. The overall entropy ratio is 0.64. The layer-wise breakdown reveals progressive sharpening:

- Shallow layers (0–1): 0.95–0.98
- Early layers (2–4): 0.70–0.80
- Mid layers (12–20): 0.57–0.66
- Deep layers (24–30): 0.48–0.57 (minimum 0.48 at layer 29)

12.1% of total attention weight falls on vision tokens overall, and within that fraction, the distribution is non-uniform in deeper layers. This progressive concentration indicates that OpenVLA has learned to selectively attend to task-relevant spatial locations, a pattern enabled by the MLP projector's preservation of per-patch spatial granularity.

Measurement conditions. All models are measured on real LIBERO rollout data: SmolVLA on 2400 frames (overall entropy) and 260 frames (per-layer analysis), π_0 on 100 frames from LIBERO-Object, OpenVLA on 250 frames from 50 episodes $\times$ 5 initial frames. Sample size differences reflect computational constraints (OpenVLA's 7B forward pass is substantially more expensive), but all measurements use task-relevant visual inputs. Entropy statistics show low variance across frames (std < 0.02 for OpenVLA, std $= 0.013$ for π_0).

M Connector Architecture and Token Independence

The six-architecture taxonomy (Table 1) spans a continuous spectrum of token independence structures, each associated with the vision-to-LLM connector rather than model scale or action head design.

Pixel-shuffle connectors create maximal entanglement. SmolVLA's 4×4 spatial folding packs 16 patches into each token by channel concatenation (CV $= 0.314$, the highest in our study). The folding is fixed and irreversible: removing one token eliminates an entire receptive field, and no downstream layer can recover the discarded content. Tokens encode non-overlapping spatial regions with heterogeneous visual content, making every token simultaneously essential and non-redundant.

Pure linear projections produce the opposite extreme. π_0 maps each SigLIP patch through a single linear layer with no cross-token interaction, yielding CV $= 0.038$ and near-maximal entropy (0.959). Tokens are genuinely interchangeable: any subset preserves the same per-patch information density as any other subset of equal size.

$\pi_0.5$ reveals a subtlety that π_0's offline-only evaluation conceals. The $\times\sqrt{d}$ learned scaling preserves near-uniform norms (CV $= 0.055$), yet the closed-loop response is strongly method-dependent: random removal at k=0.5 retains 69.4% SR while L2 norm-based removal collapses to 33.4%. The 36 pp gap arises because L2 ranking in a low-CV distribution selects spatially clustered tokens whose slight norm variations correlate with image regions, destroying the spatial coverage that this architecture requires. Token position analysis confirms this mechanism: at k=0.5, L2 selection produces a single dominant cluster containing 100% of retained tokens with only 78% spatial coverage, whereas k=0.7 yields two clusters with 97% in the largest and 100% coverage (Appendix R).

X-VLA's 12-layer BART encoder [21] introduces cross-token mixing through self-attention, creating partial dependencies that place it between the linear and MLP extremes (CV $= 0.151$, entropy $= 0.614$, the lowest among models). The resulting behavior is threshold-graded: performance holds through k=0.9 before a sharp cliff at k=0.85 (-31 pp). The cliff onset is reproducible across three seeds (std $= 0.3$ pp), confirming it as an architectural property.

MLP projectors (OpenVLA, OFT) apply per-patch non-linear transformations that create partial importance differentiation without cross-token entanglement. OpenVLA's CV (0.215) and OFT's CV (0.219) match within measurement noise, as do their entropy values (0.661 vs. 0.667); the connector, not the action head, is associated with token importance structure. CogACT [22] corroborates the graded pattern from the literature (74.2% SR under 78% pruning).

Parameter count does not predict pruning tolerance. Ordered by scale: SmolVLA (450M) $<$ X-VLA (880M) $<$ π_0 (3.3B) $<$ $\pi_0.5$ (4B) $<$ OpenVLA (7B). Ordered by tolerance, SmolVLA remains the minimum, but π_0 and $\pi_0.5$ at 3.3–4B surpass OpenVLA at 7B, and X-VLA at 880M tolerates more aggressive pruning than OpenVLA. The CV spectrum (0.038 to 0.314) tracks connector complexity rather than parameter count across a 16$\times$ scale range.

N CV as Diagnostic Metric: Extended Analysis

Token importance CV correlates with both the tolerance class of a model and the optimal pruning strategy, a correlation that emerges from a single mechanistic principle: CV measures the strength of the importance gradient that scoring methods can exploit.

Low-CV models have near-uniform L2 norms. L2 norm ranking in this regime captures spatial artifacts (slight norm variations that correlate with image position) rather than genuine importance differences. Selection based on these artifacts concentrates retained tokens on spatially correlated patches, destroying the uniform coverage that low-CV architectures depend on. Random selection preserves spatial uniformity by construction. The mechanism predicts that the random-over-L2 advantage should grow as CV decreases, and the data in Table 20 confirm this at two retention ratios. At k=0.5: $\Delta = +36.0$ pp ($\pi_0.5$, CV $= 0.055$) $\rightarrow +26.9$ pp (X-VLA, CV $= 0.151$) $\rightarrow -1.8$ pp (OpenVLA, CV $= 0.215$). At k=0.7: the same monotonic ordering holds ($+22.2$ pp $\rightarrow +6.5$ pp $\rightarrow -5.4$ pp). The pattern persists across a 2$\times$ change in absolute token budget, strengthening the interpretation from a coincidence at a single operating point to a structural property of the connector.

High-CV models present a genuine importance gradient where L2 norm ranking aligns with actual information content. OpenVLA (CV $= 0.215$) sits near the crossover: L2 outperforms random by 8.6 pp at k=0.9 where the gradient is most exploitable, but the advantage shrinks with decreasing k (5.4 pp at k=0.7, 1.8 pp at k=0.5) as the proportion of removed tokens exceeds the gradient's discriminative range.

Extreme compression introduces a partial reversal. At k=0.3, $\pi_0.5$'s L2 selection achieves 53.2 ± 3.1% SR (2 seeds) vs. random's 36.4%. When very few tokens remain, even spatially clustered high-norm tokens provide more signal than a sparse random sample. The V-shaped L2 curve is confirmed across seeds (σ=0.1 pp at the k=0.5 valley). The reversal does not invalidate the CV diagnostic; it adds a budget-dependent interaction that future work should characterize across more models.

Attention entropy provides a mechanistic supplement that resolves an apparent paradox. SmolVLA's entropy (0.952) is nearly as high as π_0's (0.959), yet SmolVLA collapses under any pruning. SmolVLA's tokens are uniformly *attended* but carry heterogeneous *content* due to pixel-shuffle folding. Entropy captures how the LLM allocates attention across tokens; CV captures the input-side importance structure. Entropy alone cannot distinguish "interchangeable" (π_0) from "entangled" (SmolVLA), but CV separates them cleanly (0.038 vs. 0.314). OpenVLA's progressive entropy sharpening from 0.98 to 0.48 across layers (Figure 8) is consistent with a graded regime where the LLM progressively concentrates on higher-importance tokens.

O Why Re-Encoding Succeeds: Extended Analysis

Learned re-encoding overcomes pixel-shuffle failure because it breaks the rigid spatial entanglement that the 4×4 folding creates. Cross-attention pooling integrates information across receptive field boundaries, redistributing content from the original 128 tokens into M compressed representations no longer tied to fixed spatial blocks.

The M=128 passthrough configuration suggests that re-encoding, not compression, is the key factor. Passthrough adds parameters without reducing token count, yet gains 13 pp over the baseline (44.8% $\rightarrow$ 58.0%). The relatively stable M-curve (M=16: 56.9% closely matches M=128: 58.0%, within 1.1 pp) shows that once entanglement is broken, token count is secondary.

A parameter-matched MLP adapter (22.14M parameters) achieves 63.4% SR at M=32, statistically indistinguishable from cross-attention ($p = 0.15$). Re-encoding through any learned mapping that integrates across token boundaries suffices; the attention mechanism itself is not essential.

The remedy is specific to the pathology. VTR succeeds precisely on architectures whose rigid spatial entanglement demands learned redistribution; for token-independent connectors where each patch already carries self-contained information, re-encoding is unnecessary.

P Offline Metrics Underestimate Closed-Loop Failure

Offline evaluation systematically underestimates pruning-induced failure because per-step averages dilute catastrophic single-step events that individually terminate episodes.

X-VLA at k=0.3 demonstrates the gap directly: offline loss is 1.0014 (near-baseline), yet closed-loop SR drops to $22.0 \pm 0.9\%$ (-77 pp). A single gripper inversion (L2 = 1.02) is a minor perturbation in average action error but terminates an episode irreversibly. π_0's offline loss at k=0.1 shows a 2.4% *improvement* over the unpruned baseline (offline metrics only; no closed-loop evaluation available), a result that closed-loop evaluation may contradict if similar catastrophic events occur at deployment frequency.

$\pi_0.5$ adds a distinct form of evidence: the 36 pp gap between L2 and random removal at k=0.5 is invisible to any offline metric that averages across token subsets or selection methods. Offline evaluation would report degradation for both methods; only closed-loop evaluation reveals that the *choice* of pruning strategy, not just the pruning ratio, is associated with whether the model remains functional.

Q Confounder Analysis

The six architectures span three connector families but differ simultaneously in LLM backbone (PaliGemma for π_0/$\pi_0.5$, Llama-2 for OpenVLA, Qwen2 for SmolVLA), preventing clean isolation of the connector effect from backbone differences. The π_0–OpenVLA comparison (same 256 SigLIP-based tokens, different connector, backbone, and encoder fusion strategy) provides the closest available approximation, but a matched-backbone experiment remains the ideal control. This comparison involves differences in connector, backbone, and encoder fusion, limiting causal interpretation. Token count also varies across models (50 for X-VLA, 128 for SmolVLA, 256 for π_0/OpenVLA, 768 for $\pi_0.5$); the π_0–OpenVLA pair controls for this confounder within the CV analysis while still exhibiting the predicted CV-gap relationship.

The spatial clustering hypothesis for L2 failure on low-CV models is confirmed by token position analysis (Appendix R).

OpenVLA L2 and random results use different seeds (seed42 and seed7 respectively); same-seed (seed42) random replication confirms the cross-seed pattern at k=0.3 (21.4% random vs. 25.0% L2, gap 3.6 pp) and k=0.7 (44.4% random vs. 49.8% L2, gap 5.4 pp). Same-seed (seed42) random at k=0.5 yields 35.8% (L2: 39.2%, gap 3.4 pp), consistent with the cross-seed comparison in Table 20.

R Token Position Analysis

To validate the spatial clustering hypothesis for L2 norm failure on low-CV models, we analyze the spatial distribution of tokens selected by L2 norm ranking on $\pi_0.5$ across three retention ratios.

At k=0.7 (538 of 768 tokens retained): DBSCAN clustering identifies 2 spatial clusters, with 97% of retained tokens in the largest cluster and 100% spatial coverage of the original image grid. The broad coverage explains why L2 selection still achieves 55.0% SR at this retention ratio.

At k=0.5 (384 tokens retained): a single dominant cluster contains 100% of retained tokens with only 78% spatial coverage. The complete concentration into one spatial region confirms the clustering hypothesis: L2 norm ranking selects tokens from a contiguous high-norm region while abandoning 22% of the spatial field. This is the retention ratio where L2 performance reaches its minimum (33.4% SR, the V-shape valley).

At k=0.3 (230 tokens retained): 3 clusters emerge, with 86% in the largest and 78% spatial coverage. The fragmentation into multiple clusters at extreme compression is consistent with the partial L2 recovery observed at this ratio ($53.2 \pm 3.1\%$ SR): when the budget is very small, L2 selection is forced to sample from multiple spatial regions, partially restoring coverage.

These results directly confirm that L2 norm ranking on low-CV models introduces spatial bias that degrades performance, and that the severity peaks at intermediate retention ratios where concentration is maximal.

S CV Scaling Law: Full Data

Table 20: L2 norm vs. random removal at two retention ratios. The gap Δ(R$-$L2) scales monotonically with CV at both k=0.5 and k=0.7: low-CV models require random selection (uniform importance $\rightarrow$ L2 degenerates to spatial clustering), while high-CV models tolerate either strategy. The monotonic ordering holds across retention ratios despite a 2$\times$ change in token budget.

Model	CV	Tokens	L2 SR (%)	Random SR (%)	Δ(R$-$L2)
k=0.5 (50% retention)					
π_0.5	0.055	768	33.4	**69.4**	+36.0
X-VLA	0.151	50	43.4±0.8	**70.3**±0.9	+26.9
OpenVLA	0.215	256	**39.2**	37.4	−1.8
k=0.7 (70% retention)					
π_0.5	0.055	768	55.0	**77.2**	+22.2
X-VLA	0.151	50	58.6±0.4	**65.1**±0.1	+6.5
OpenVLA	0.215	256	**49.8**	44.4	−5.4

T CV-Strategy Interaction at Extreme Compression

The monotonic CV-gap relationship (Table 20) holds at k=0.5 and k=0.7, but partially reverses at k=0.3 for π_0.5. At this extreme budget, L2 norm selection achieves $53.2 \pm 3.1\%$ SR (2 seeds) vs. random's 36.4%, favoring L2 by $\sim$17 pp. The reversal is consistent across both seeds (seed42: 51.0%, seed43: 55.4%; Table 8).

The mechanism is intuitive: when very few tokens remain (k=0.3 retains 230 of 768 tokens), even spatially clustered high-norm tokens provide more visual signal than a sparse random sample. At intermediate budgets (k=0.5, 384 tokens), random selection preserves enough spatial coverage to outperform L2's clustered selection; at extreme budgets, coverage from random selection becomes too sparse to be useful, and the concentrated signal from L2 selection dominates.

The interaction implies that the optimal pruning strategy depends on both CV and the absolute token budget. For low-CV models, the crossover point between random-preferred and L2-preferred regimes lies between k=0.3 and k=0.5. Characterizing this crossover across additional models and budgets is left to future work.

U Inference Cost-Benefit Analysis

The primary deployment benefit of Smart Compress at this model scale is memory, not latency. Reducing from 128 to 32 vision tokens shrinks the KV-cache footprint for the vision portion by 75%, a meaningful reduction for memory-constrained robotic hardware where KV-cache size can limit the number of concurrent policy instances or maximum context length.

End-to-end inference latency improves only modestly: 154.78 ms at M=32 versus 164.33 ms at full token count, a $1.06\times$ speedup (6% reduction). At M=64: 156.16 ms ($1.05\times$). The gap between token reduction and latency gain reflects SmolVLA's compact size (451.5M parameters): self-attention over 128 vision tokens is not the inference bottleneck, and the compressor's 22.2M-parameter cross-attention computation partially offsets downstream savings. Token pruning becomes a more compelling latency optimization for VLAs with larger token budgets (>256 tokens) or larger backbones (7B+ parameters), where attention cost scales quadratically with sequence length. Whether the training cost (100K steps, 6.5 hours per configuration) is justified depends on deployment context: for VLAs deployed across many tasks, the one-time training cost is amortized; for rapid prototyping, retraining is a barrier.

Comparison with the official checkpoint. Smart Compress at $M = 32$ ($60.1 \pm 1.3\%$ over 3 seeds) recovers from 0% SR under hard pruning to performance comparable to the official SmolVLA checkpoint (59.2%, CI [54.8%, 63.4%]). However, the difference is not statistically significant (two-sample t-test, $p \approx 0.72$), and the comparison should be interpreted cautiously given different training conditions (batch size 4 vs. 64, single vs. 8 GPUs). The primary contribution of VTR is recovering functional performance where all deletion methods yield 0%, not matching the official baseline.